Lecture 13: Moment Generating Functions

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1 Moment generating functions

Moment generating functions (MGFs) are functions associated with a random variable. They encode a *lot* of information about the distribution. They are also called **transforms**.

Like the PDF/PMF and CDF, these functions can uniquely determine the distribution of a random variable.

1.1 Definition

i Definition (Moment generating function)

The **moment generating function (MGF)** associated with a random variable X is

$$M_X(s) = \mathbb{E}\left[e^{sX}\right] = \begin{cases} \sum_x e^{sx}\, p_X(x) & X \text{ discrete} \\ \int_{-\infty}^{\infty} e^{sx}\, f_X(x)\, dx & X \text{ continuous} \end{cases}$$

Note that the MGF is a *function* of a parameter s; taking $\mathbb{E}[e^{sX}]$ transforms a function (the PDF/PMF) into another function. This is unlike how we use the PDF/PMF where we think of the specific numerical outcome of the function at a specific value of x (i.e., an outcome of X).

1.1.1 Example: a simple PMF

Let X have PMF,

$$p_X(x) = \begin{cases} \frac{1}{8}, & x = -1 \\ \frac{5}{8}, & x = 2 \\ \frac{1}{4}, & x = 3 \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$M_X(s) = \sum_{x \in \{-1,2,3\}} p_X(x) e^{sx} = \frac{1}{8}e^{-s} + \frac{5}{8}e^{2s} + \frac{1}{4}e^{3s}.$$

1.1.2 Example: Poisson MGF

Let $X \sim \text{Poisson}(\lambda)$.

Then

$$M_X(s) = \sum_{x=0}^{\infty} e^{sx}\, e^{-\lambda} \frac{\lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^s)^x}{x!} = e^{-\lambda} e^{\lambda e^s} = e^{\lambda(e^s - 1)}.$$

1.1.3 Example: standard normal RV

Let $Z \sim N(0,1)$.

Then

$$M_Z(s) = \int_{-\infty}^{\infty} e^{sz} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz.$$

Complete the square: $sz - \frac{z^2}{2} = -\frac{(z-s)^2}{2} + \frac{s^2}{2}$.

Then

$$M_Z(s) = e^{s^2/2} \underbrace{\int \frac{1}{\sqrt{2\pi}} e^{-(z-s)^2/2}\, dz}_{=1} = e^{s^2/2}.$$

1.1.4 Exercise: exponential RV

Now, let's find the MGF of an exponential random variable!

> **Exercise**
>
> Let $X \sim \text{Exponential}(\lambda)$. So, $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Find $M_X(s)$. For which s does it exist?

Solution

$$M_X(s) = \int_0^{\infty} e^{sx} \lambda e^{-\lambda x}\, dx = \lambda \int_0^{\infty} e^{(s-\lambda)x}\, dx = \frac{\lambda}{s-\lambda} e^{(s-\lambda)x} \Big|_{x=0}^{x=\infty}.$$

For $s \geq \lambda$ the integral diverges. For $s < \lambda$, this is

$$-\frac{\lambda}{s-\lambda} = \frac{\lambda}{\lambda - s}.$$

The MGF is only defined for $s < \lambda$.

1.2 MGF of a linear function of an RV

Fact

If $Y = aX + b$, then

$$M_Y(s) = e^{sb}\, M_X(as).$$

Proof:

$$M_Y(s) = \mathbb{E}\left[e^{s(aX+b)}\right] = e^{sb}\,\mathbb{E}\left[e^{(sa)X}\right] = e^{sb} M_X(as)$$

1.2.1 Exercise: arbitrary normal RV

Exercise

Let $X \sim N(\mu, \sigma^2)$. Find $M_X(s)$.
Hint: $X = \sigma Z + \mu$ with $Z \sim N(0, 1)$.

Solution

$$M_X(s) = e^{\mu s} M_Z(\sigma s) = e^{\mu s + \sigma^2 s^2/2}$$

1.3 Why "moment generating"?

Theorem

Differentiating the moment generating function of X with respect to s and evaluating at $s = 0$ produces the moments of X,

$$\frac{d^n}{ds^n} M_X(s)\Big|_{s=0} = \mathbb{E}[X^n]$$

Let's look at a continuous random variable (the calculation for a discrete RV is the same but with a summation instead of an integral). Note that

$$\frac{d}{ds} M_X(s) = \frac{d}{ds}\int_{-\infty}^{\infty} e^{sx} f_X(x)dx = \int_{-\infty}^{\infty} \frac{d}{ds} e^{sx} f_X(x)dx = \int_{-\infty}^{\infty} x e^{sx} f_X(x)dx.$$

Now, evaluate at $s = 0$,

$$\frac{d}{ds} M_X(s) \mid_{s=0} = \int_{-\infty}^{\infty} x f_X(x)dx = \mathbb{E}[X].$$

The general statement is obtained by differentiating n times, then evaluating at $s = 0$.

1.3.1 Example: check on the simple PMF

First moment (mean). Differentiate term by term:

$$M_X'(s) = -1 * \frac{1}{8} e^{-s} + 2 * \frac{5}{8} e^{2s} + 3 * \frac{1}{4} e^{3s}.$$

Evaluating at $s = 0$:

$$\mathbb{E}[X] = M_X'(0) = -\frac{1}{8} + 2 * \frac{5}{8} + 3 * \frac{2}{8} = \frac{15}{8}.$$

Second moment. Differentiate again:

$$M_X''(s) = (-1)^2 * \frac{1}{8}e^{-s} + (2^2) * \frac{5}{8}e^{2s} + (3^2) * \frac{1}{4}e^{3s}.$$

Evaluating at $s = 0$:

$$\mathbb{E}[X^2] = M_X''(0) = \frac{1}{8} + \frac{20}{8} + \frac{18}{8} = \frac{39}{8}.$$

Check against the definition:

$$\mathbb{E}[X] = (-1)\tfrac{1}{8} + (2)\tfrac{5}{8} + (3)\tfrac{2}{8} = \tfrac{15}{8}, \qquad \mathbb{E}[X^2] = (1)\tfrac{1}{8} + (4)\tfrac{5}{8} + (9)\tfrac{2}{8} = \tfrac{39}{8}.$$

1.3.2 Example: exponential moments

Recall that if $X \sim \text{Exponential}(\lambda)$ then $M_X(s) = \dfrac{\lambda}{\lambda - s}$.

Then $M_X'(s) = \frac{\lambda}{(\lambda - s)^2}$ and evaluating at $s = 0$ yields

$$\mathbb{E}[X] = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

Also, $M_X''(s) = \frac{2\lambda}{(\lambda - s)^3}$ and evaluating at $s = 0$ yields

$$\mathbb{E}[X^2] = \frac{2\lambda}{\lambda^3} = \frac{2}{\lambda^2}.$$

1.3.3 Exercise: Bernoulli RV

> **Exercise**
>
> Let $X = 1$ with probability p and $X = 0$ otherwise. Find $M_X(s)$, then use it to compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

Solution

$M_X(s) = (1-p)e^{0*s} + pe^{1*s} = 1 - p + pe^s$, so $M_X'(0) = M_X''(0) = p$. Hence $\mathbb{E}[X] = \mathbb{E}[X^2] = p$ and $\text{Var}(X) = p(1-p)$.

> **Bernoulli moments**
>
> If $X \sim \text{Bernoulli}(p)$, then every moment equals p:
>
> $$E[X^k] = p \quad \text{for all } k \geq 1.$$
>
> We already knew this from the direct calculation, since $X \in \{0, 1\}$ means $X^k = X$:
>
> $$E[X^k] = 0^k(1-p) + 1^k\, p = p.$$
>
> The MGF agrees. With $M_X(s) = (1-p) + pe^s$, every derivative is $M_X^{(k)}(s) = pe^s$, so $M_X^{(k)}(0) = p$.

1.4 MGF facts

Facts

1. $M_X(0) = 1$.
2. If $X \geq 0$, then $M_X(s) \to \mathbb{P}(X = 0)$ as $s \to -\infty$.

Fact 1. Recall $M_X(s) = \mathbb{E}[g_s(X)]$ for some function g_s. What function $g_s(x)$ are we taking the expectation of when $s = 0$? Why does this force $M_X(0) = 1$?

Fact 2. Let's test it on two nonnegative random variables.

- Let $U \sim \text{Uniform}(0, 1)$. Compute $M_U(s)$, then find $\lim_{s\to-\infty} M_U(s)$. Does it match $\mathbb{P}(U = 0)$?
- Let $X \sim \text{Poisson}(\lambda)$. Use $M_X(s) = e^{\lambda(e^s-1)}$ and find $\lim_{s\to-\infty} M_X(s)$. Does it match $\mathbb{P}(X = 0)$?

1.4.1 Fact 2

If $X \geq 0$, then $M_X(s) \to \mathbb{P}(X = 0)$ as $s \to -\infty$.

Think about it: What happens to $g_s(x) = e^{sx}$ as $s \to -\infty$ when $x = 0$? When $x > 0$? How does that explain Fact 2?

As $s \to -\infty$, $e^{sx} \to 1$ if $x = 0$ and $e^{sx} \to 0$ if $x > 0$. So $g_s(X)$ converges to the indicator $\mathbf{1}\{X = 0\}$, and its expectation converges to $\mathbb{P}(X = 0)$.

1.5 Inversion property

Fact (Inversion Property)

The MGF $M_X(s)$ associated with a random variable X uniquely determines the CDF of X, assuming that $M_X(s)$ is finite for all s in some interval $[-a, a]$ where a is a positive number.

Exact inversion formulas that allow us to recover the PMF/PDF from an MGF exist, but they are hard to use.

In practice, we typically **pattern-match** against known MGFs.

1.5.1 Example: inverting the simple MGF

$$M_X(s) = \tfrac{1}{8}e^{-s} + \tfrac{5}{8}e^{2s} + \tfrac{2}{8}e^{3s}$$

The coefficient of e^{sx} is $\mathbb{P}(X = x)$:

$$p_X(-1) = \tfrac{1}{8}, \qquad p_X(2) = \tfrac{5}{8}, \qquad p_X(3) = \tfrac{2}{8}$$

1.5.2 Example: inverting a geometric MGF

We are told that the MGF associated with a random variable X is

$$M_X(s) = \frac{\frac{1}{4}e^s}{1 - \frac{3}{4}e^s}.$$

Recall that $\frac{1}{1-\alpha} = 1 + \alpha + \alpha^2 + \cdots$, so

$$\begin{aligned} M_X(s) &= \tfrac{1}{4}e^s(1 + \tfrac{3}{4}e^s + (\tfrac{3}{4})^2e^{2s} + \cdots) \\ &= \tfrac{1}{4}(\tfrac{3}{4})^0e^s + \tfrac{1}{4}(\tfrac{3}{4})^1e^{2s} + \tfrac{1}{4}(\tfrac{3}{4})^2e^{3s} + \cdots \\ &= \sum_{k=1}^{\infty} p(1-p)^{k-1}e^{ks} \end{aligned}$$

where $p = 1/4$.

X is a geometric random variable with parameter $p = 1/4$!

1.6 Sums of independent RVs

Addition of independent random variables corresponds to multiplication of the associated moment generating functions:

Theorem

If X and Y are independent, then $M_{X+Y}(s) = M_X(s)\, M_Y(s)$.

$$M_{X+Y}(s) = \mathbb{E}\left[e^{s(X+Y)}\right] = \mathbb{E}\left[e^{sX}e^{sY}\right] = \mathbb{E}\left[e^{sX}\right]\mathbb{E}\left[e^{sY}\right]$$

1.6.1 Example: sum of independent Poissons

If $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ are independent, then

$$M_{X+Y}(s) = e^{\lambda(e^s-1)}\, e^{\mu(e^s-1)} = e^{(\lambda+\mu)(e^s-1)}.$$

So $X + Y \sim \text{Poisson}(\lambda + \mu)$. We derived this previously in the homework using convolution! The MGF is an alternative route to determining distribution and parameters of sums of independent random variables.

1.7 Activity

Divide into groups of four. Two in the group will work on (1) and two will work on (2). Then the groups of two will share out to one another.

Activity

Derive the MGF of each of the following.

1. Binomial(n, p)
2. Uniform(a, b)

Answers

1. Binomial is a sum of n independent Bernoulli(p) random variables, so the MGF is the product of their MGFs: $(1 - p + pe^s)^n$
2. Note that $f_X(x) = \frac{1}{b-a}$, so $\mathbb{E}[e^{sX}] = \int_a^b \frac{e^{sx}}{b-a}dx = \frac{e^{sx}}{s(b-a)}\,\big|_{x=a}^{x=b} = \dfrac{e^{sb} - e^{sa}}{s(b-a)}$ for $s \neq 0$ and 1 at $s = 0$.

1.8 MGF of a joint distribution

Definition

Consider n jointly distributed random variables $X_1, X_2, \cdots, X_n$. The associated **multivariate moment generating function** is

$$M_{X_1,\cdots,X_n}(s_1,\cdots,s_n) = \mathbb{E}\left[e^{s_1X_1+\cdots+s_nX_n}\right]$$

Derivatives at $(0,\cdots,0)$ give joint moments, e.g.,

$$\left.\frac{\partial^2 M_{X_1,X_2}}{\partial s_1\,\partial s_2}\right|_{(0,0)} = \mathbb{E}[X_1X_2].$$

The inversion property extends to the multivariate case. Suppose $Y_1,\cdots,Y_n$ is another set of random variables and we have $M_{Y_1,\cdots,Y_n}(s_1,\cdots,s_n) = M_{X_1,\cdots,X_n}(s_1,\cdots,s_n)$ for all $(s_1,\cdots,s_n)$ belonging to some n-dimensional hypercube with positive volume, then the joint distribution of $X_1,\cdots,X_n$ is the same as the joint distribution of $Y_1,\cdots,Y_n$.